

>, <, =]

Hello

Selection
Sort

Bubble
Sort

Insertion
Sort

Sorting

eg:- [20, 15, 32, 7, 19]

⇒ [7, 15, 19, 20, 32]

→ Selectⁿ
→ Bubble
→ Insert

→ Merge
→ Quick } Recursion

① Selection Sort

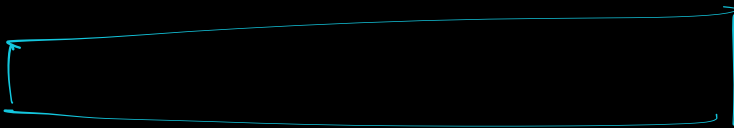
eg:- [7, 5, 3, 2, 6]

0	1	2	3	4
2	3	5	6	7

Sorted

0	1	2	3	4
2	3	5	6	7
✓	✓			

① Searching for Minimum Element



② Swap with its respective index

```

5 function selectionSort(arr){
6   let n = arr.length;
7   for(let i=0; i<n-1; i++){
8     let idx = i;
9     let minIdx = i;
10    for(let j=i; j<n; j++){
11      if(arr[j] < arr[minIdx]){
12        minIdx = j;
13      }
14    }
15
16    // swap idx - minIdx
17    let temp = arr[idx];
18    arr[idx] = arr[minIdx];
19    arr[minIdx] = temp;
20  }
21 }

```

① Minimum

$\min = 2$

$idx = 0$

swap (arr[0], arr[3])

[2, 5, 3, 7, 6]

$idx = 1$

j = 1 [5, 3, 7, 6]

min = 3 (2)
↑
index

swap (arr[idx], arr[j])

[2, 3, 5, 7, 6]

$idx = 2$

min = 5 (2)

swap (arr[idx], arr[j])

[2, 3, 5, 7, 6]

$$\text{idx} = 3$$

$$\begin{matrix} 0 & 1 & 2 & 3 & 4 \\ [2 & 3 & 5 & 7 & 6] \end{matrix}$$

$$\text{min} = 6(4)$$

$$\text{swap}(\text{arr}[\text{idx}], \text{arr}[4])$$

$$\boxed{2 \quad 3 \quad 5 \quad 6 \quad 7}$$

$$\text{idx} = 4$$

eg:- $\begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \\ [60, 42, 31, \underline{7}, 12, 15] \end{matrix}$

①

$$\underline{\text{idx} = 0}$$

$$\text{min} = 7$$

$$\underline{\text{idx of min elem} = 3}$$

$$\text{swap}(\text{arr}[\text{idx}], \text{arr}[\quad])$$

$$[7, 42, 31, 60, 12, 15]$$

$$(2) \text{ idx} = 0$$

$$\text{min} = 12$$

$$\text{elem idx} = 1$$

$$\left[\begin{array}{c} 0 \\ 1 \end{array} \right] \left| \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 42 & 31 & 60 & 12 & 15 \end{array} \right]$$

$$\text{swap}(\text{arr}[\text{idx}], \text{arr}[\text{elem idx}])$$

$$\left[\begin{array}{c} 0 \\ 1 \end{array} \right] \left| \begin{array}{ccccc} 2 & 3 & 4 & 5 \\ 31 & 60 & 42 & 15 \end{array} \right]$$

$$(3) \text{ idx} = 1$$

$$\text{min} = 15$$

$$\text{elem idx} = 5$$

$$\text{swap}(\text{arr}[\text{idx}], \text{arr}[\text{elem idx}])$$

$$\left[\begin{array}{c} 0 \\ 1 \\ 2 \end{array} \right] \left| \begin{array}{ccccc} 3 & 4 & 5 \\ 7 & 12 & 15 & 60 & 42 & 31 \end{array} \right]$$

$$(4) \text{ idx} = 2$$

$$\text{min} = 31$$

$$\text{elem idx} = 5$$

$$\text{swap}(\text{arr}[\text{idx}], \text{arr}[\text{elem idx}])$$

$$\left[\begin{array}{c} 0 \\ 1 \\ 2 \\ 3 \end{array} \right] \left| \begin{array}{ccccc} 4 & 5 \\ 7 & 12 & 15 & 31 & 42 & 60 \end{array} \right]$$

$$(5) \text{ idx} = 3$$

$$\text{min} = 42$$

$$\text{elem idx} = 4$$

$$\text{swap}(\text{arr}[\text{idx}], \text{arr}[\text{elem idx}])$$

$$\left[\begin{array}{c} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \right] \left| \begin{array}{ccccc} 5 \\ 7 & 12 & 15 & 31 & 42 & 60 \end{array} \right]$$

TC of selection

$$TC = \frac{n^2}{2} + \frac{n}{2}$$

$$\frac{n^2}{2}$$

$$\underline{\underline{O(n^2)}}$$

$i=0 \rightarrow$

$$\left. \begin{array}{c} n \\ n-1 \\ n-2 \\ n-3 \\ \vdots \\ 2 \\ 1 \end{array} \right\} +$$

Selection Sort

$$TC = O(n^2)$$

$$SC = O(1)$$

$$\frac{n(n+1)}{2}$$

* Problem

eg:-

0	1	2	3	4	5	
6	2	7	2	3	1	5

target = 5

	0	1	2	3	4	5
	1	2	3	5	6	27

eg:-

0	1	2	3	4	5
7	9	4	0	10	1

 = arr

$O(n)$ → solution



target = 7

0	1	2	3	4	5
0	1	4	7	9	10

* Problem - 2

Smallest greater Alt

arr =

13	6	17	12
----	---	----	----

ans = $17 \begin{pmatrix} 12 & 1 & -10 \end{pmatrix} \quad 13$

1 $\rightarrow$ 17

$$2 \rightarrow 17, 12$$


12 $\rightarrow$ 13, 17

y!-

0	1	2	3	4
15	7	20	3	10 12
↑	↑			

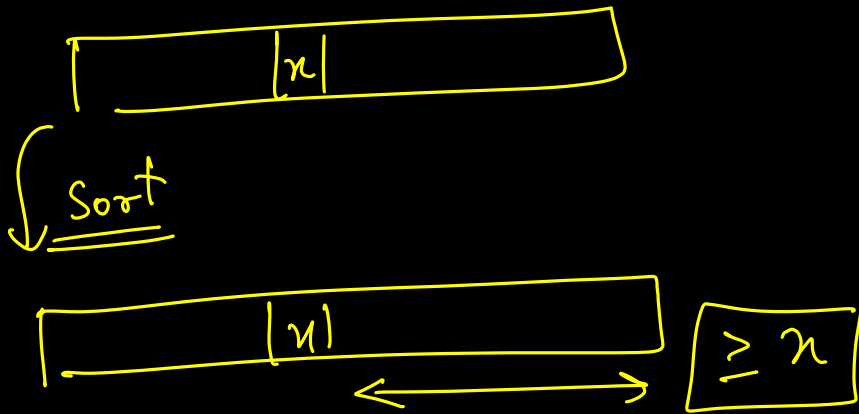
20	10	$-\infty$	7	12	15
----	----	-----------	---	----	----

$$15 \rightarrow \{ \underline{20} \}$$
$$7 \rightarrow \left\{ \frac{15, 20}{10, 12} \right\}$$

20 →

$$3 \rightarrow \{15, 7, 20, 10, 12\}$$
$$10 \rightarrow \{12, 15, 20\}$$
$$12 \rightarrow \{15, 20\}$$

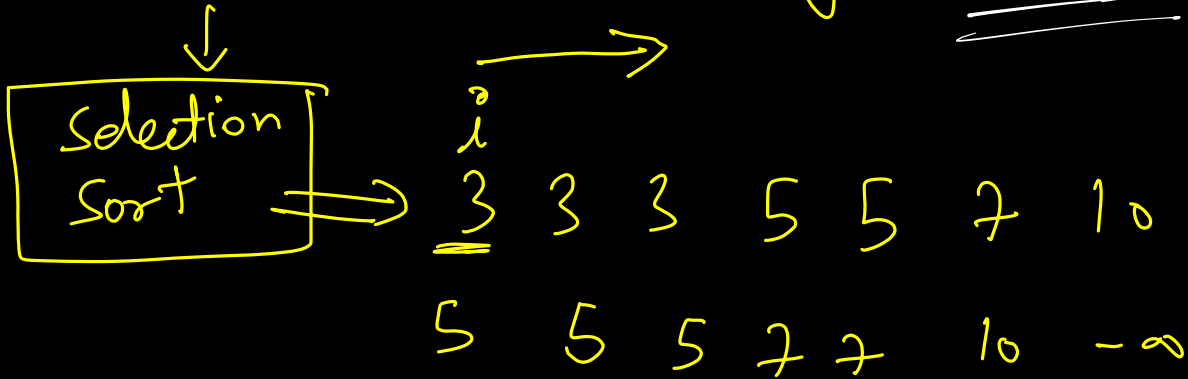
Sort



15 7 2 3 10 12

<u>3</u>	<u>7</u>	<u>10</u>	12	15	20
7	10	12	15	20	$-\infty$

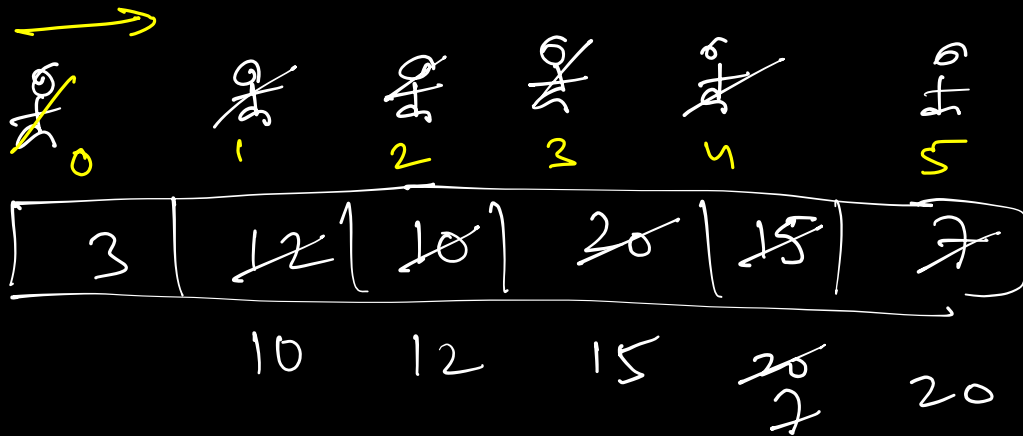
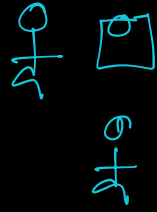
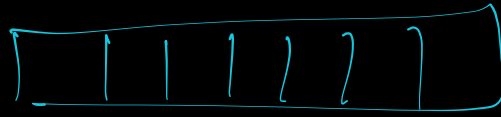
① Sort the given array (HW)



$$\text{curr} = \text{arr}[i] = 3$$

$$\underline{j = i + 1 \rightarrow n} \Rightarrow \underline{\text{arr}[j]} \neq \text{arr}[i]$$

Bubble Sort



① Always start from 0 index.

② if my current element is greater than the next element then only swap

$$\text{curr} > \text{next ele}$$

③ Irrespective of swap move-on to next position

n-1 ^(for outer loop) times j = my curr position

```
for (j=0 ; j < n-1 ; j++) {  
    cur = arr[j]  
    next = arr[j+1]  
    if (cur > next)  
        swap  
}  
}
```

eg:-

0	1	2	3	4
4	7	2	0	9

Dry Run using

Bubble Sort.

~~| | | | | |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
| 4 | 7 | 2 | 0 | 9 |~~

① curr = 4
next = 7
(curr → next) X

curr = 7
next = 2

curr = 7
next = 0

curr = 7
next = 9

curr

0	1	2	3	4
4	2	0	7	9

② swaps

② ~~4~~⁹ ~~7~~² ~~2~~⁰ ~~0~~⁷ 9

curr = 4
next = 2) ✓

curr = 4
next = 0) ✓

curr = 4
next = 7) X

curr = 7
next = 9)

② swaps

2 0 4 7 9

③

0	1	2	3	4
2	0	4	7	9

curr = 2
next = 0) ✓

curr = 2
next = 4) X

① swap

0	1	2	3	4
0	2	4	7	9

$$\begin{aligned}\text{Total swaps done} &= 2 + 2 + 1 \\ &= 5\end{aligned}$$

eg:-

5 | 3 | 0 | 1 | 10 | 7 | 9

no. of swaps while sorting this using bubble sort.

7 | 7

5	→	(3, 0, 1)	→	3	}	7
3	→	(0, 1)	→	2		
0	→	x	→	0		
1	→	x	→	0		
10	→	(7, 9)	→	2		
7	→	x	→	0		

$$\underline{\underline{Q)}} \quad [7 \mid 5^* \mid 10 \mid 20 \mid 0 \mid 5]$$

no. of swaps while doing bubble sort.

$$\cancel{5} \mid \check{8}$$

$\cancel{6}$

$$7 \rightarrow (5, 0, 5) \rightarrow 3$$

$$5^* \rightarrow (0) \rightarrow 1$$

$$10 \rightarrow (0, 5) \rightarrow 2$$

$$20 \rightarrow (0, 5) \rightarrow 2$$

$$0 \rightarrow x \rightarrow 0$$

$$5 \rightarrow x \rightarrow 0$$

$$\underline{\underline{8}}$$

Insertion Sort

eg:-

0	1	2	3	4	5	6
2	7	9	1	10	12	8

 key = 17

Insert (key = 17) at index = 2

0	1	2	3	4	5	6	7
2	7	17	9	1	10	12	8

0	1	2	3	4	5	6	7
2	7	9	7	10	12	8	8
			↑	9	1	10	12

17 $arr[i+1] = arr[i]$

$i = 6$ { $arr[i+1] = arr[i]$
 $arr[7] = arr[6]$

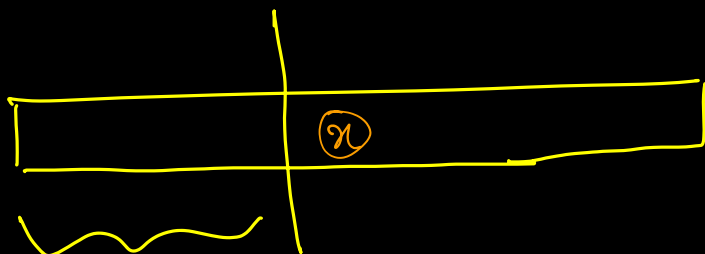
$i = 5$ { $arr[6] = arr[5]$

$i = 4$ { $arr[5] = arr[4]$

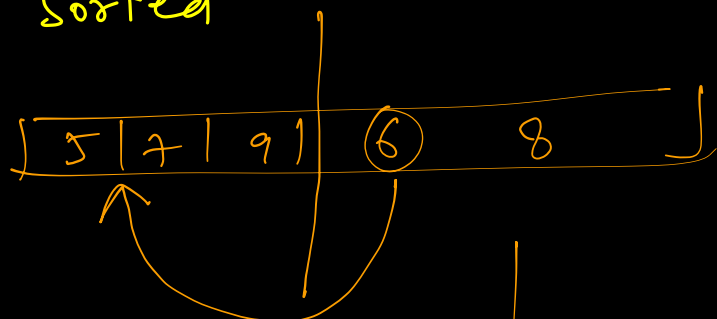
$i = 3$ { $arr[4] = arr[3]$

$i = 2$ { $arr[3] = arr[2]$

$arr[i] = key$



Sorted



5 6 7 9 | 8

5 6 7 8 9 | 11

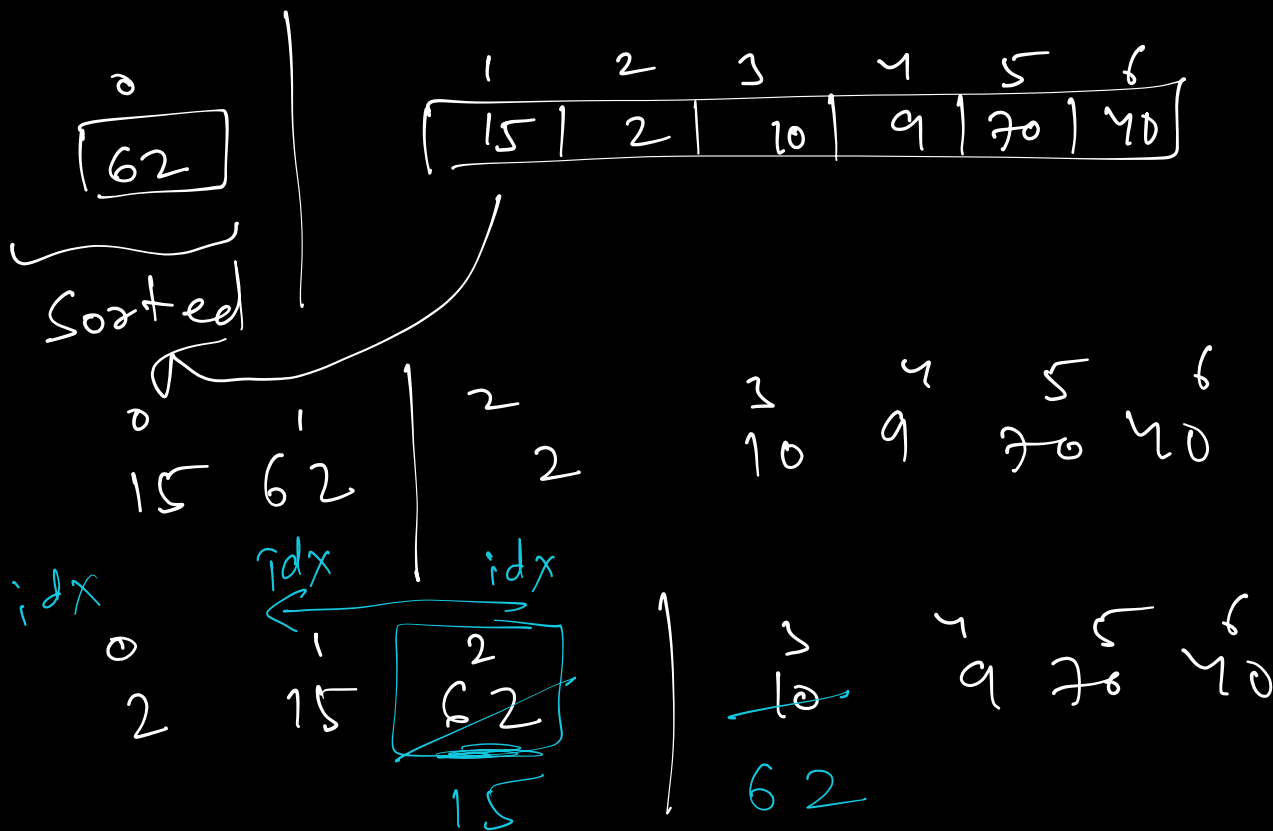
5 6 7 8 9 11

2

2 5 6 7 8 9 11

eg:-

0	1	2	3	4	5	6
62	15	2	10	9	70	40

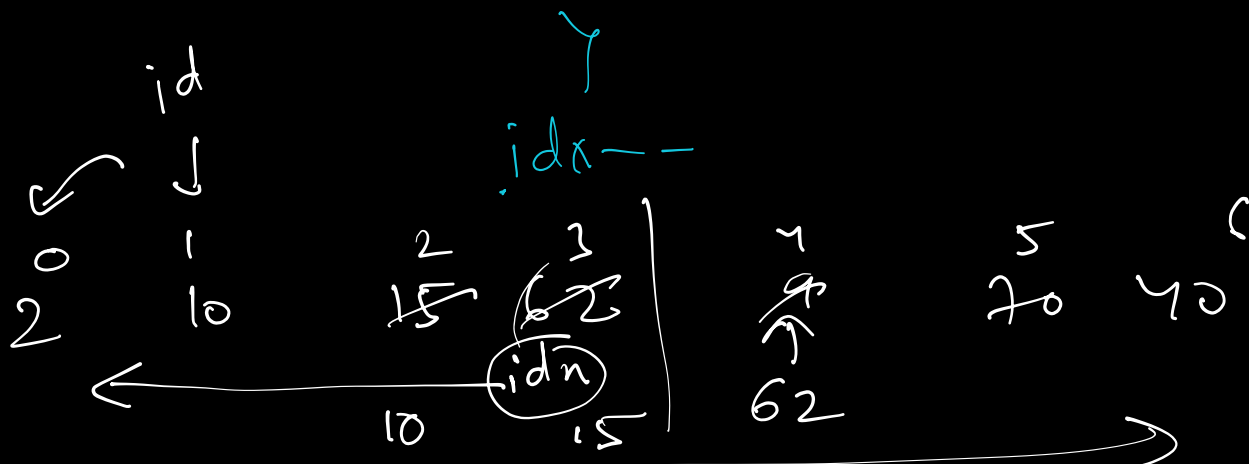


key = 10

(idx → 0)

$arr[idx] = key$

if ($arr[idx] > key$) {
 $arr[idx+1] = arr[idx]$



①

key = 9

```

for ( idx = i-1 ; idx ≥ 0 ; idx-- )
    if ( arr[idx] > key )
        arr[idx+1] = arr[idx]

```

}
 }

arr[idx+1] = key

idx = 3
 key = 9

0	1	2	3	4
			idx	
2	10	15	62	9
		10	15	62

arr[idx] = 62 > 9

arr[idx+1] = arr[idx]

arr[4] = arr[3]

idx = 2

arr[idx] > key (15 > 9)

arr[idx+1] = arr[idx]

arr[3] = arr[2]

idx = 1

arr[idx] > key (10 > 9)

arr[idx+1] = arr[idx]

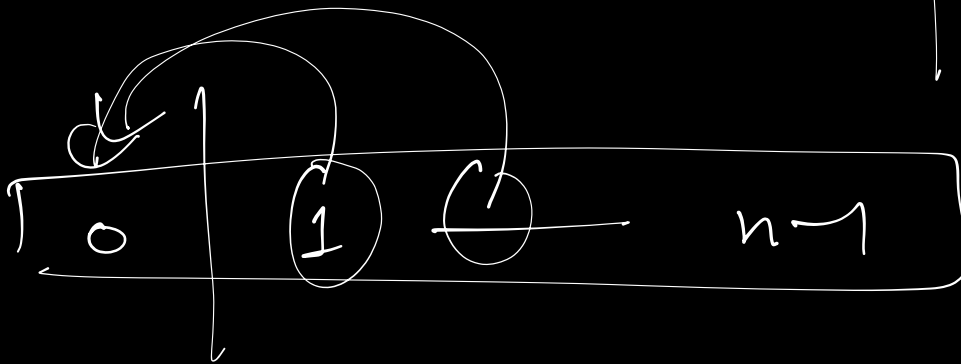
arr[2] = arr[1]

idx = 0

arr[idx] > key (2 > 9)

$$\boxed{\text{arr}[\text{idx} + 1] = \text{key}}$$

2 9 10 15 62 | 76 90



```

for( i = 1 → n-1 ) {
    key = arr[i], idx
    for( idx = i-1 ; idx ≥ 0 ; idx-- )
        if( arr[idx] > key ) {
            arr[idx+1] = arr[idx]
        }
        else { break; }
}

```

using
while

arr[idx+1] = key

* TC of Insertion Sort

$$f = \frac{n(n+1)}{2} = O(n^2)$$

* SC of insertion sort

$$f = O(1)$$

Inbuilt Sort

```
arr.sort();
```

"10"
"2"
"2" > "10"

```
arr = [1, 10, 2]
```

```
arr.sort();
```

```
console.log(arr); → o/p = ?
```

$\text{arr.sort}((a, b) \Rightarrow a - b)$ $\rightarrow$ ascending order
 $\Downarrow$
Comparator

$O(n \log n)$

$((a, b) \Rightarrow b - a) \Rightarrow$ for
Descending